Vivek Gopalakrishnan

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Research Overview

I am a 5th year PhD candidate in Medical Engineering and Medical Physics at the Harvard-MIT Program in Health Sciences and Technology. My research centers on *patient-specific machine learning*, developing personalized, physics-informed computer vision models capable of extracting 3D information from 2D imaging. I am particularly interested in applications in diagnostics, image-guided interventions, and surgical robotics.

Research Areas: *computer vision*, *medical physics*, *surgical robotics*, *interventional imaging*, *X-ray*.

Education

Massachusetts Institute of Technology — *2021 - Present*
Ph.D. in Medical Engineering and Medical Physics — Advisor: Polina Golland

Johns Hopkins University — *2020 - 2021*
M.S.E. in Biomedical Engineering — Advisors: Joshua Vogelstein and Carey Priebe

Johns Hopkins University — *2017 - 2021*
B.S. in Biomedical Engineering — Advisor: Joshua Vogelstein

Research Experience

Computer Science and Artificial Intelligence Laboratory, MIT — *2021 - Present*

- PhD candidate in the Medical Vision Group building intelligent intraoperative image guidance systems.
- Developed `DiffDRR`, an open-source differentiable X-ray renderer with over 50,000 downloads [W1].
- Developed `xvr`, the first 2D/3D registration algorithm to achieve consistent sub-mm accuracy [C1, P1].
- Building 3D vision systems for neurosurgery [J8, J9], orthopedics [J5, A1], and radiotherapy [C2].
- Collaborating with surgical robotics startups to integrate these algorithms into FDA-approved devices.

Department of Biomedical Engineering, Johns Hopkins University — *2018 - 2021*

- Undergraduate researcher in the NeuroData Lab studying statistical graph theory for neuroscience.
- Developed novel machine learning algorithms to analyze populations of graph-valued data [J4, J10].
- Applied these methods to discover neuroconnectively similar subtypes of autism spectrum disorder from multi-subject connectomics data [J3].

Fellowships and Awards

Rising Star in AI, University of Michigan — 2025
MIT HEALS Graduate Fellowship (*9 months of full funding*), MIT — 2025
Best Poster Award, MIT-MGB AI Cures Conference — 2024
Neuroimaging Training Program Grant (*12 months of full funding*), MIT — 2024
President's Innovation Award, Society of Biomolecular Imaging and Informatics — 2023
Takeda Fellowship (*9 months of full funding*), MIT — 2023
Neuroimaging Training Program Grant (*12 months of full funding*), MIT — 2022
Joseph C. Pistritto Research Fellowship, Johns Hopkins University — 2019
Provost's Undergraduate Research Award, Johns Hopkins University — 2020

Publications

Conferences

C2 **Vivek Gopalakrishnan**, Neel Dey, and Polina Golland. "PolyPose: Deformable 2D/3D Registration via Polyrigid Transforms". *Neural Information Processing Systems (NeurIPS)*, 2025 (*to appear*).

C1 **Vivek Gopalakrishnan**, Neel Dey, and Polina Golland. "Intraoperative 2D/3D Image Registration via Differentiable X-ray Rendering". *Computer Vision and Pattern Recognition (CVPR)*, 2024.
🏆 **Best Poster Award** at the MIT-MGB AI Cures Conference.

Journals

J10 **Vivek Gopalakrishnan**, Jaewon Chung, Eric Bridgeford, Benjamin Pedigo, Jesús Arroyo, Lucy Upchurch, G. Allan Johnson, Nian Wang, Youngser Park, Carey Priebe, and Joshua Vogelstein. "Multiscale Comparative Connectomics". *Imaging Neuroscience*, 2025.

J9 Sarah Frisken, **Vivek Gopalakrishnan**, David Chlorogiannis, Nazim Haouchine, Alexandre Cafaro, Alexandra Golby, William Wells, and Rose Du. "Spatiotemporally Constrained 3D Reconstruction from Biplanar Digital Subtraction Angiography". *International Journal of Computer Assisted Radiography and Surgery*, 2025.

J8 Charles Downs, Matthijs van der Sluijs, Sandra A.P. Cornelissen, Frank te Nijenhuis, Robert van Oostenbrugge, Wim H. van Zwam, **Vivek Gopalakrishnan**, Xucong Zhang, Ruisheng Su, and Theo van Walsum. "Improving Automatic Cerebral 3D-2D CTA-DSA Registration". *International Journal of Computer Assisted Radiology and Surgery*, 2025.

J7 Joshua Marchant, Natalie Ferris, Diana Grass, Magdelena Allen, **Vivek Gopalakrishnan**, Mark Olchanyi, ..., Hui Wang, Anastasia Yendiki, Susie Huang, Bruce Rosen, Randy Gollub. "Mesoscale Brain Mapping: Bridging Scales and Modalities in Neuroimaging". *Neuroinformatics*, 2024.

J6 Sarah Frisken, Nazim Haouchine, David Chlorogiannis, **Vivek Gopalakrishnan**, Alexandre Cafaro, William Wells, Alexandra Golby, Rose Du. "VESCL: An Open-Source 2D Vessel Contouring Library". *International Journal of Computer Assisted Radiography and Surgery*, 2024.

J5 Andrew Abumoussa, **Vivek Gopalakrishnan**, Benjamin Succop, Michael Galgano, Sivakumar Jaikumar, Yueh Lee, and Deb Bhowmick. "Machine Learning for Automated and Real-Time 2D-3D Registration of the Spine Using Only a Single Radiograph". *Neurosurgical Focus*, 2023.

J4 Jaewon Chung, Eric Bridgeford, Jesus Arroyo, Benjamin Pedigo, Ali Saad-Eldin, **Vivek Gopalakrishnan**, Liang Xiang, Carey Priebe, and Joshua Vogelstein. "Statistical Connectomics". *Annual Review of Statistics and Its Application*, 2021.

J3 Nian Wang, Robert Anderson, David Ashbrook, **Vivek Gopalakrishnan**, Youngser Park, Carey Priebe, Yi Qi, Rick Laoprasert, Joshua Vogelstein, Robert Williams, and G. Allan Johnson. "Variability and Heritability of Mouse Brain Structure: Microscopic MRI Atlases and Connectomes for Diverse Strains". *NeuroImage*, 2020.

J2 **Vivek Gopalakrishnan**, Eliezer Bose, Usha Nair, Yuwei Cheng, and Musie Ghebremichael. "Pre-HAART CD4+ T-Lymphocytes as Biomarkers of Post-HAART Immune Recovery in HIV-Infected Children with or without TB Co-Infection". *BMC Infectious Diseases*, 2020.

J1 Jong Soo Lee, Elijah Paintsil, **Vivek Gopalakrishnan**, and Musie Ghebremichael. "A Comparison of Machine Learning Techniques for Classification of HIV Patients with Antiretroviral Therapy-Induced Mitochondrial Toxicity". *BMC Medical Research Methodology*, 2019.

Preprints

P1 **Vivek Gopalakrishnan**, Neel Dey, David Chlorogiannis, Andrew Abumoussa, Darren B. Orbach, Sarah Frisken, and Polina Golland. "Rapid Patient-Specific Neural Networks for Intraoperative X-ray to Volume Registration". *arXiv*, 2025 (*Under Review*).

Peer-Reviewed Workshops

W5 Noe Bertramo, Gabriel Duguey, and **Vivek Gopalakrishnan**. "DiffUS: Differentiable Ultrasound Rendering from Volumetric Imaging". *Advances in Simplifying Medical Ultrasound Workshop at MICCAI*, 2025.

W4 Federica Facente, Benjamin Billot, **Vivek Gopalakrishnan**, Manasi Kattel, Wen Wei, Polina Golland, Hervé Delingette, Nicholas Ayache, and Pierre Berthet-Rayne. "Multi-stage CNN for Fast Registration of 3D Preoperative CTs to 2D Intraoperative X-rays". *Collaborative Intelligence and Autonomy in Image-guided Surgery Workshop at MICCAI*, 2025.

W3 Mohammadhossein Momeni*, **Vivek Gopalakrishnan***, Neel Dey, Polina Golland, and Sarah Frisken. "Differentiable Voxel-based X-ray Rendering Improves Sparse-View 3D CBCT Reconstruction". *Machine Learning and the Physical Sciences Workshop at NeurIPS*, 2024.

W2 **Vivek Gopalakrishnan**, Jingzhe Ma, and Zhiyong Xie. "Grad-CAMO: Learning Interpretable Single-Cell Morphological Profiles from 3D Cell Painting Images". *Computer Vision for Microscopy Image Analysis Workshop at CVPR*, 2024.
🏆 **President's Innovation Award** from the Society of Biomolecular Imaging and Informatics.

W1 **Vivek Gopalakrishnan** and Polina Golland. "Fast Auto-Differentiable Digitally Reconstructed Radiographs for Solving Inverse Problems in Intraoperative Imaging". *Clinical Image-based Procedures Workshop at MICCAI*, 2022.

Abstracts

A1 Michael Hachadorian and **Vivek Gopalakrishnan**. "Recovery of 3D Component Position in Reverse Shoulder Arthroplasty from Postoperative Radiographs via 2D/3D Registration". *American Shoulder and Elbow Surgeons Fellows' Symposium*, 2025.

Invited Talks

Patient-Specific Machine Learning for 3D Intraoperative Image Guidance

- *University of Michigan | AI Symposium*, Ann Arbor, MI 2025
- *United Imaging Intelligence*, Boston, MA 2025
- *Noah Medical*, San Francisco, CA 2025
- *Indian Institute of Science*, Bangalore, India 2025
- *GE HealthCare AI*, Bangalore, India 2025
- *Boston Medical Imaging Workshop*, MIT, Cambridge, MA 2024

Differentiable X-ray Rendering for Fast Intraoperative 2D/3D Image Registration

- *GE HealthCare AI*, Bangalore, India 2024
- *MIT Visual Computing Seminar*, Cambridge, MA 2024

Learning Interpretable Single-Cell Morphological Profiles from 3D Cell Painting Images

- *Computer Vision for Microscopy Image Analysis Workshop (CVPR 2024)*, Seattle, WA 2024
- *Society of Biomolecular Imaging and Informatics*, Boston, MA 2023

Fast Auto-Differentiable DRRs for Intraoperative Imaging Problems

- *Boston Medical Imaging Workshop*, Brigham and Women's Hospital, Boston, MA 2022
- *Image-Guided Neurosurgery Meeting*, Boston, MA 2022
- *Medical Image Computing and Computer Assisted Interventions CLIP*, Singapore 2022

Multiscale Comparative Connectomics

- *Presentations by Undergraduates in Life Sciences and Engineering*, Baltimore, MD 2021
- *PathAI*, Online 2021
- *Neuromatch Conference*, Online 2020

Mentorship

Research

- Zengtian Deng (PhD), UCLA — Summer 2025
  Co-supervised with Drs. Bibo Shi and Tao Zhao, Noah Medical
  Project: Implementing **xvr** [P1] in a deployed robotic surgical system
- Noe Bertramo (Masters) and Gabriel Duguey (Masters) — Summer 2025
  Project: Developing a differentiable ultrasound rendering engine [W4]
- Michelle Wu (Undergrad), MIT — Summer 2025 & Fall 2026
  Co-supervised with Professor Polina Golland, MIT
  Project: A fully automated system for postoperative evaluation of shoulder replacement surgery
- Jonathan Tjandra (Undergrad), MIT — Spring 2025 & Fall 2026
  Co-supervised with Professor Polina Golland, MIT
  Project: Learning to identifying lung nodules from chest radiographs from synthetic data
- Erik Xie (Undergrad), MIT — Fall 2024 & Spring 2025
  Co-supervised with Professor Polina Golland, MIT
  Project: Custom CUDA kernels for accelerated X-ray rendering
- Hossein Momeni (Undergrad), Truman State University — Summer 2024 & Fall 2024
  Co-supervised with Professor Sarah Frisken, Harvard Medical School
  Project: Differentiable X-ray rendering improves Sparse-view 3D CBCT reconstruction [W3]
  Current position: PhD student at UC Berkeley (EECS)

Community

- Graduate Resident Advisor, McCormick Hall, MIT — 2024 - Present

Teaching

Advances in Computer Vision (Graduate) — Spring 2025
Head Teaching Assistant, Massachusetts Institute of Technology

NeuroData Design II (Undergraduate/Graduate) — Spring 2021
Head Teaching Assistant, Johns Hopkins University

NeuroData Design I (Undergraduate/Graduate) — Fall 2020
Head Teaching Assistant, Johns Hopkins University

Computational Cardiology Laboratory (Undergraduate) — Fall 2020
Teaching Assistant, Johns Hopkins University

Linear Algebra (Undergraduate) — Fall 2020
Teaching Assistant, Johns Hopkins University

Service

PhD Admissions Committee, Harvard-MIT Health Sciences and Technology — 2025
Reviewer, Medical Image Analysis (MEDIA) — 2025
Reviewer, IEEE Transactions Pattern Analysis and Machine Intelligence — 2025
Reviewer, IEEE Transactions on Medical Imaging (🏆 *Distinguished Reviewer*) — 2024
Reviewer, Journal of Imaging Informatics in Medicine — 2024
Reviewer, Information Processing in Medical Imaging (IPMI) — 2025
Reviewer, Medical Imaging Computing and Computer Assisted Interventions (MICCAI) — 2024, 2025
Reviewer, Medical Imaging with Deep Learning (MIDL) — 2024

Software

`xvr`: Training patient-specific 2D/3D registration models in 5 min 2025
Primary author and maintainer (https://github.com/eigenvivek/xvr/)

`DiffDRR`: GPU-accelerated & differentiable X-ray rendering in PyTorch 2022
Primary author and maintainer (https://github.com/eigenvivek/DiffDRR/)

Work Experience

Consultant (Medical Imaging Machine Learning Scientist), Noah Medical *2025 - Present*